

SUTTSU, Japan

WMO: 474210

Lat: 42.7956N

Lon: 140.2244E

Elev: 115

StdP: 14.63

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.6	18.7	5.9	7.5	19.5	8.1	8.4	21.5	25.4	30.4	23.0	30.4	11.0	310	0.522

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.7	80.7	73.2	78.1	71.5	76.0	70.3	74.5	78.9	73.0	76.9	71.5	74.9	7.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.1	123.5	77.3	71.6	117.2	75.9	70.2	111.6	74.5	38.3	79.1	36.9	77.1	35.5	75.0	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.7	19.9	17.6	DB	13.3	85.9	2.9	3.2	11.2	88.3	9.5	90.1	7.9	92.0	5.7	94.3
			WB	11.8	76.9	2.8	2.1	9.8	78.4	8.1	79.7	6.6	80.9	4.5	82.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	27.4	28.2	33.9	43.6	52.9	60.0	67.7	70.8	64.7	53.7	41.6	31.4
	DBStd	15.98	4.79	5.12	5.22	5.72	5.36	4.69	4.48	4.10	4.68	5.46	7.02	5.58
	HDD50	2916	701	611	499	207	29	1	0	0	0	27	263	578
	HDD65	6522	1166	1031	964	641	375	162	22	4	61	351	702	1043
	CDD50	2222	0	0	0	16	120	300	548	644	442	141	11	0
	CDD65	353	0	0	0	0	1	11	106	182	53	0	0	0
	CDH74	966	0	0	0	1	10	28	260	576	91	0	0	0
	CDH80	110	0	0	0	0	0	2	33	71	4	0	0	0
Wind	WSAvg	8.3	9.2	9.3	8.7	8.6	8.9	8.2	7.7	7.1	6.8	7.3	8.2	9.1
Precipitation	PrecAvg	49.8	4.9	3.5	2.7	2.4	2.6	2.4	3.8	5.1	5.9	5.1	5.9	5.5
	PrecMax	66.1	7.8	6.8	8.4	4.6	5.9	5.3	9.8	11.3	12.8	9.0	10.1	8.2
	PrecMin	37.1	2.3	2.0	0.7	0.5	0.5	0.3	0.9	1.6	2.2	2.3	3.0	3.0
	PrecStd	7.0	1.3	1.1	1.5	1.2	1.3	1.4	2.2	2.5	2.9	1.9	1.9	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.7	43.6	52.1	67.0	74.9	77.7	84.0	85.6	80.1	69.4	61.0	49.8
		MCWB	37.1	38.8	43.7	53.0	59.1	65.6	74.5	75.9	72.9	61.3	53.9	45.9
	2%	DB	38.2	40.2	47.2	60.5	69.2	73.2	79.6	81.8	76.5	66.3	57.5	45.3
		MCWB	34.9	36.0	40.8	50.0	56.8	63.6	72.2	74.3	69.9	59.9	52.3	41.7
	5%	DB	36.2	37.7	44.2	56.2	64.9	70.2	76.7	79.1	74.2	64.0	54.9	42.1
		MCWB	32.9	34.0	39.0	47.2	54.5	62.9	71.1	72.7	68.6	58.4	50.3	38.2
	10%	DB	34.3	35.8	41.7	52.7	61.6	67.4	74.3	77.0	72.2	62.1	52.3	39.5
		MCWB	30.9	32.3	37.2	45.5	53.1	61.5	69.4	71.5	66.9	57.0	47.8	35.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.8	39.4	44.7	55.2	61.6	67.8	75.6	77.4	74.3	63.7	56.0	46.7
		MCDB	39.8	42.6	50.0	63.5	71.5	73.9	81.0	83.4	78.1	66.3	59.6	49.1
	2%	WB	35.5	36.6	41.8	51.1	58.6	65.6	73.6	75.3	72.1	61.4	53.3	42.2
		MCDB	37.7	39.4	45.9	58.5	66.4	70.6	78.2	79.9	75.0	64.7	56.4	45.0
	5%	WB	33.4	34.6	39.7	48.5	56.4	63.9	71.9	73.9	70.0	59.6	50.7	38.7
		MCDB	35.7	37.3	43.7	54.4	62.5	68.4	75.6	77.9	72.9	63.2	54.4	41.6
	10%	WB	31.3	32.5	37.5	46.3	54.6	62.3	70.3	72.5	67.8	57.7	48.2	36.0
		MCDB	34.0	35.5	41.2	51.7	59.9	66.7	73.7	76.2	71.3	61.5	52.0	39.0
Mean Daily Temperature Range	5% DB	MDBR	6.9	7.5	8.9	11.8	12.5	11.0	9.7	9.7	11.1	11.3	9.6	7.5
		MCDBR	9.3	10.6	13.2	18.2	19.1	15.7	13.3	12.3	12.2	12.9	13.5	11.0
		MCWBR	8.5	9.3	9.9	10.7	10.4	8.8	7.3	6.8	7.6	8.8	11.0	10.1
	5% WB	MCDBR	9.6	10.5	12.6	15.9	16.3	13.5	11.3	10.8	9.9	11.3	13.1	10.9
		MCWBR	9.3	9.7	10.3	10.2	9.7	8.1	6.5	6.4	7.2	9.0	11.8	10.7
Clear-Sky Solar Irradiance	taub	0.359	0.409	0.476	0.535	0.552	0.527	0.512	0.474	0.420	0.414	0.400	0.378	
	taud	2.049	1.913	1.828	1.777	1.827	1.973	2.088	2.213	2.311	2.170	2.104	2.031	
	Ebn at Noon	232	238	237	235	235	242	245	250	256	237	215	209	
	Edh at Noon	35	48	59	67	65	57	50	43	36	36	33	33	
All-Sky Solar Radiation	RadAvg	365	596	1031	1432	1595	1533	1365	1369	1194	851	466	309	
	RadStd	29	63	89	144	144	157	159	144	113	123	52	49	24

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page